

Agentic AI Lifestyle Companion

Gen AI Hackathon Challenge

Aligned with Vodacom's AI-first strategy, let's bring the AI Lifestyle Companion to life through this challenge.

Build an Agentic AI companion that supports voice and text and helps customers plan for their needs, discover relevant products, choose suitable options and complete a simulated purchase.

Use customer profiles, past interactions, the product catalogue and recommender model output to personalize the experience across Home, Shop and Rewards.

The companion should

1. **Understand customer needs:** Identify their goal, budget and preferences. Ask follow-up questions when needed.
2. **Recommend suitable choices:** Find relevant products, explain recommendations and help customers compare options.
3. **Engage proactively and cross-sell:** Use available data to start helpful conversations and suggest complementary products when relevant. Teams decide when and how.
4. **Adapt to feedback:** Update suggestions when customers change their needs or reject an option.
5. **Support a simulated purchase:** Use the supplied functions (AI_Companion_Starter) to manage the basket, show the checkout summary and create an order after customer confirmation.

Demo customer identification

The first page must include a text box to enter a user_id. This replaces login for the simulation. If the ID is invalid, ask for a valid one.

Data and resources

File or resource	What it includes	How to use it
train Customer profiles	Customer ID, region, age band, household size, membership, device, budget, quality preference, language, marketing opt-in, owned items and interests.	Personalize the experience. Link to interactions and recommender output using user_id.
products Product catalogue	Product IDs, names, domains, categories, brands, prices, discounts, quality, style tags, descriptions, availability, OS compatibility, repeatability and subscription flags.	Find and compare products, check eligibility and calculate discounted prices. Link to interactions and recommender output using product_id.
interactions Customer activity history	Event ID, customer and product IDs, app section, event type/date, browsing duration, session depth, related events and transaction details where available.	Understand behavior across Home, Shop and Rewards. Support personalization, proactive engagement and cross-selling. Join train on user_id and products on product_id.
Team-defined filename Recommender model output	Ranked product recommendations for each customer.	Use recommendations alongside the customer's current request, preferences and constraints.
AI_Companion_Starter.zip Simulation starter package	Working basket and simulated-order code, sample CSVs, tool schemas, setup instructions, demo and tests. Function descriptions are in AI_Companion_Functions.xlsx.	Extract the package and connect its functions to your companion. The code and sample-data files are inside this ZIP.

Deliverables

Deliverable	What to submit
1. Agent design diagram	A flow diagram showing the agent(s), tools, data sources, memory and how they work together.
2. Source code	The complete code with dependencies and clear instructions to run the solution.
3. Chatbot link	A working HTTPS link hosted in the provided AWS account.
4. Screen recording	A recording demonstrating the test scenarios shared by the organizers 30 minutes before the submission deadline. Show the conversation, actions and results.

Evaluation criteria

Criterion	Weight	What judges will assess
Understanding and personalization	25%	Understands the customer's goal and uses their profile, history, catalogue and recommender output to provide relevant choices.
Agent design and tool execution	20%	Clear workflow, appropriate tool use, correct basket actions and successful checkout after customer confirmation.
Memory and context management	20%	Retains relevant details, handles references to earlier options, updates changed preferences, manages longer conversations and separates customers.
Proactive engagement and cross-selling	15%	Uses data to offer helpful suggestions and complementary products that fit the customer's needs and budget.
Voice, text and usability	10%	Clear, easy-to-use experience with consistent context across voice and text.
Accuracy and reliability	10%	Uses correct product facts and prices, respects eligibility, and handles missing information, unavailable choices and errors clearly.

Example test scenarios

- "I'm preparing my new home. Can you help me choose what I need?"
- "I would like to buy a new smartphone. What would suit me?"
- "That's too expensive. My budget is now 100, and I don't want the first option."
- After seeing the checkout summary: "Remove that item before placing the order."
- "Find something in this category within my budget."